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Taighde Éireann
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ViAd-NET : Frame-level billboard classification

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Table of content

- Background
- Proposed Method
- Experimental Result
- Conclusion

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Background

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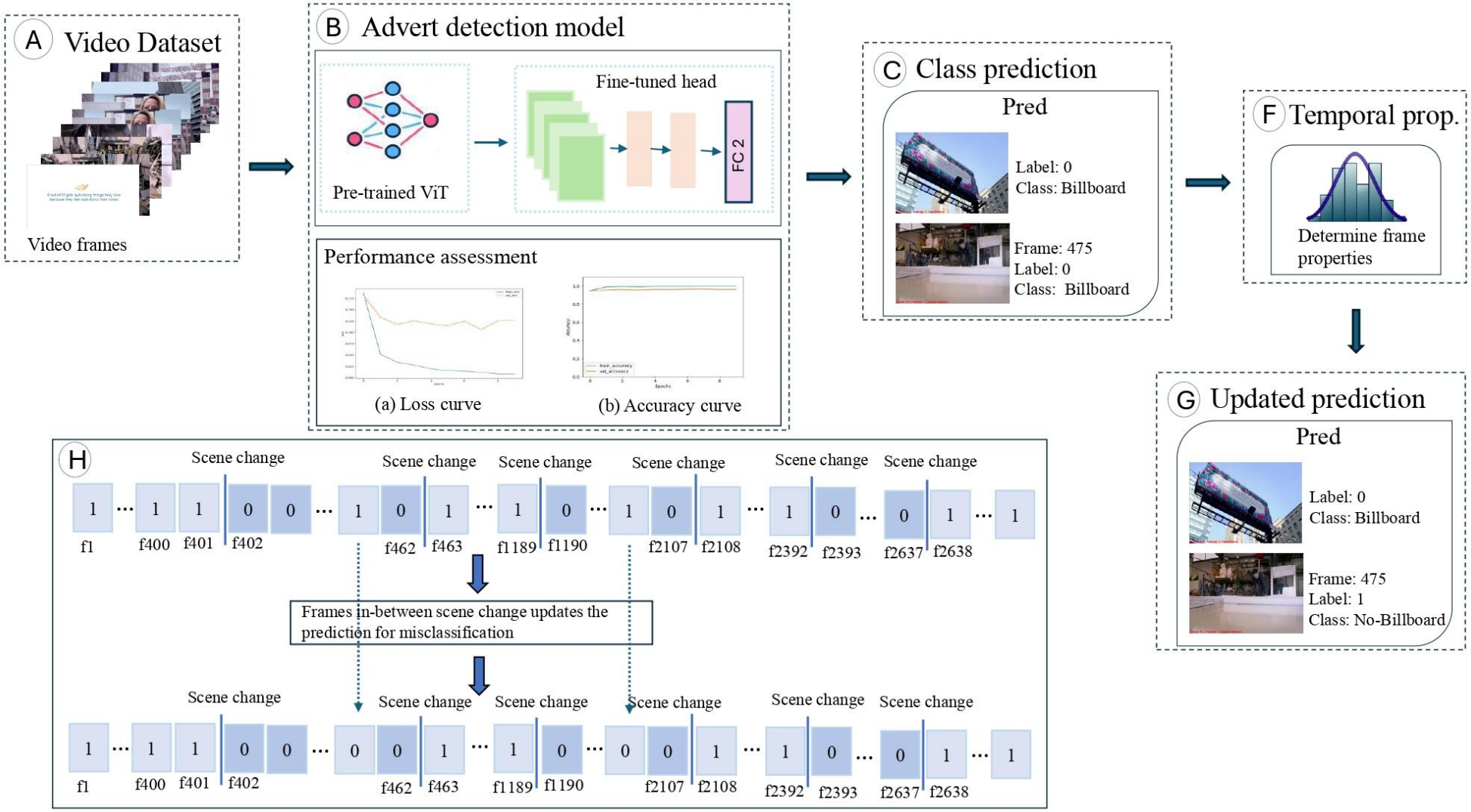
- Emphasizing advert placement, integrating new ads into images and video frames
- Traditional methods of advert detection is time-consuming.

Current state-of-art:

- ➢ Detection of billboards in streamed soccer sports videos by detecting ground edges using hue slicing and analyzing edge pixels through the Hough transform [1]
- ➢ Poor model performance in detecting billboards [2,3]

1. R. F. Rahmat, D. Dennis, O. S. Sitompul, S. Purnamawati, and R. Budiarto, “Advertisement billboard detection and geotagging system with inductive transfer learning in deep convolutional neural network”, Telkomnika (Telecommunication Computing Electronics and Control), 17(5):2659 - 2666, 2019.
2. S. Chavan, D. Kerr, S. Coleman, and H. Khader,“Billboard Detection in the Wild”, In Irish Machine Vision and Image Processing Conference Proceedings-DCU, Ireland, pp. 57-64, 2021. Irish Pattern Recognition and Classification Society.
3. K. Bochkarev and E. Smirnov, “Detecting advertising on building facades with computer vision,” Procedia Computer Science, vol.156, pp. 338–346, 2019.

Proposed method



Experimental results #1

Videos	Class prediction				Updated prediction			
	Acc	Prec	Rec	F1-score	Acc	Prec	Rec	F1-score
V1	47.1%	0.50	0.24	0.33	87.0%	1	0.80	0.88
V2	99.0%	0	0	0	99.0%	0	0	0
V3	92.0%	0	0	0	92.6%	1	0.92	0.95
V4	81.1%	0.87	0.48	0.62	85.3%	0.96	0.84	0.89

Frame-level prediction of correctly classified class instances (both billboard and no-billboard scenes)

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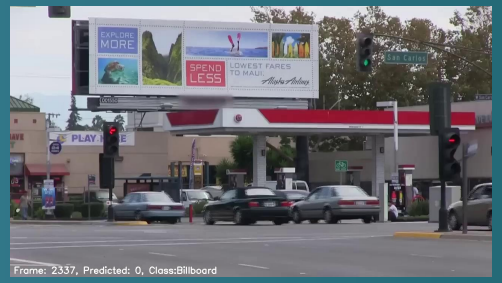
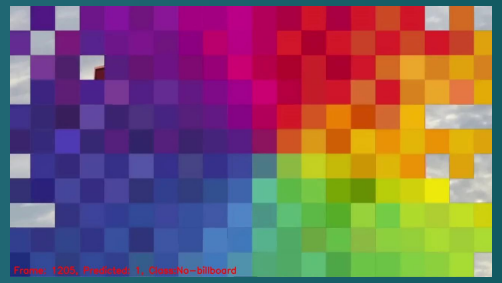
Video 1

Video 4





Frame-level prediction of correctly classified class instances

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Video 2
(only no-billboard instances)

Video 3
(only billboard instances)

Conclusion

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- ❑ ViAd-Net model: Billboard classification with temporal properties to update false negatives in between scene changes.
- ❑ **Performance**: Achieves high accuracy ranging from 85% to 99% across test scenarios.
- ❑ **Real-World Potential**: A robust solution for applications in digital advertising and multimedia content analysis

Key applications:

- ❑ **Digital Advertising**: Automatic detection and classification of billboards in video content for ad tracking, dynamic placement, and campaign performance monitoring.
- ❑ **Multimedia**: Facilitates content-based indexing and retrieval by classifying and annotating billboard appearances across video databases.

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Thank You